

LAB 10

ANSWERS FOR PRACTICE QUESTIONS.

1) Find a matrix $\diamond$ whose eigenvalues are 75 and 15 and eigenvectors are $[1,1]^T$ and $[1, -1]^T$. Also use diagonalization to find $\diamond^5$.

```
clc
d1 = 75;
d2 = 15;
Ev1 = [1,1]';
Ev2 = [1,-1]';
% Create the eigenvector matrix
V = [Ev1,Ev2];
% Create the diagonal matrix from the eigenvalues
D = diag([d1,d2]);
A = V * D * inv(V)
```

```
A = 2×2
    45    30
    30    45
```

```
% Compute A^5 using the diagonalization
D5 = D^5;
A5 = V * D5 * inv(V)
```

```
A5 = 2×2
109 ×
    1.1869    1.1861
    1.1861    1.1869
```

2) Generate a matrix with eigenvalues as 3,5,7,11 and eigenvectors as $[1,1,2,2]$, $[1,0,1,0]$, $[2,1,2,0]$ and $[0,1,1,0]$.

```
clc
Ld1 = 3;
Ld2 = 5;
Ld3 = 7;
Ld4 = 11;
Ev1 = [1;1;2;2];
Ev2 = [1;0;1;0];
Ev3 = [2;1;2;0];
Ev4 = [0;1;1;0];
V = [Ev1,Ev2,Ev3,Ev4]
```

```
V = 4×4
     1     1     2     0
     1     0     1     1
     2     1     2     1
     2     0     0     0
```

```
D = diag([Ld1,Ld2,Ld3,Ld4])
```

```
D = 4x4
    3     0     0     0
    0     5     0     0
    0     0     7     0
    0     0     0    11
```

```
A = V*D*inv(V)
```

```
A = 4x4
    9     4    -4    -1
   -4     7     4    -4
   -2     4     7    -5
    0     0     0     3
```

3) Find the eigenvalues of $A A^T$ and $A^T A$ and verify both matrices are similar if $A = \begin{bmatrix} 1 & 3 & 0 & 4 \\ 1 & 6 & 2 & 7 \end{bmatrix}$.

```
clc
A = [1 3; 0 4; 1 6; 2 7];
AA_T = A * A'
```

```
AA_T = 4x4
    10    12    19    23
    12    16    24    28
    19    24    37    44
    23    28    44    53
```

```
A_TA = A' * A
```

```
A_TA = 2x2
     6    23
    23   110
```

```
eig_AA_T = eig(AA_T)
```

```
eig_AA_T = 4x1
   -0.0000
    0.0000
    1.1405
   114.8595
```

```
eig_A_TA = eig(A_TA)
```

```
eig_A_TA = 2x1
    1.1405
   114.8595
```

```
non_zero_AA_T = sort(eig_AA_T(eig_AA_T > 1e-10));
non_zero_A_TA = sort(eig_A_TA(eig_A_TA > 1e-10));
% Check if eigenvalues are same
if all(abs(non_zero_AA_T - non_zero_A_TA) < 1e-10)
    disp('Eigenvalues of AA^T and A^T A are the same.');
```

```
else
    disp('Eigenvalues of AA^T and A^T A are different.');
```

```
end
```

4) Consider the matrix, $A = \begin{bmatrix} -4 & 1 & 1 \\ 1 & -6 & 7 \\ 1 & 7 & -9 \end{bmatrix}$. Find two matrices B and C that are similar to matrix A and verify their eigenvalues are same.

```
clc
A = [-4, 1, 1; 1, -6, 7; 1, 7, -9];
% Generate random invertible matrix M (Two matrices A and B are similar if there
exists an invertible matrix M)
M = randi([1, 9], 3, 3);
% Create similar matrices B and C
B = inv(M) * A * M;
C = M * A * inv(M);
eig_A = sort(eig(A))
```

```
eig_A = 3×1
-14.6610
-4.4761
0.1371
```

```
eig_B = sort(eig(B))
```

```
eig_B = 3×1
-14.6610
-4.4761
0.1371
```

```
eig_C = sort(eig(C))
```

```
eig_C = 3×1
-14.6610
-4.4761
0.1371
```

```
% Check if the eigenvalues are identical for all matrices
if norm(eig_A - eig_B) < 1e-10 && norm(eig_A - eig_C) < 1e-10
    disp('Eigenvalues of A, B, and C are the same.');
```

```
else
    disp('Eigenvalues of A, B, and C are different.');
```

```
end
```

Eigenvalues of A, B, and C are the same.

5) Create a 10×10 non-diagonal matrix with eigenvalues as 1, 2, ..., 10.

[Hint: Create a similar matrix using a non-singular matrix B:

$B = B * \text{diag}([1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10]) * B^{-1}$)

```
clc
% Define a random non-singular matrix B
B = rand(10)
```

```
B = 10×10
0.6843    0.5568    0.2413    0.9548    0.5862    0.2273    0.8594    0.5068 ...
0.6568    0.7120    0.6229    0.9196    0.9258    0.0486    0.7811    0.9508
```

0.9904	0.4879	0.5229	0.3848	0.5751	0.1692	0.2038	0.3946
0.0337	0.6176	0.4132	0.1626	0.0100	0.2584	0.9933	0.5845
0.4243	0.2138	0.2178	0.7968	0.8094	0.1979	0.0936	0.6065
0.4900	0.6457	0.8586	0.1138	0.6088	0.6057	0.6506	0.7146
0.5835	0.3806	0.8610	0.1588	0.4799	0.8237	0.2152	0.4015
0.0833	0.1037	0.2839	0.3558	0.2684	0.8106	0.2439	0.8587
0.6602	0.3775	0.6154	0.8478	0.2581	0.8022	0.3397	0.9205
0.0523	0.2629	0.7795	0.5828	0.4810	0.7081	0.1979	0.7508

```
% Define a diagonal matrix with eigenvalues 1 to 10
```

```
D = diag(1:10)
```

```
D = 10x10
```

1	0	0	0	0	0	0	0	0	0
0	2	0	0	0	0	0	0	0	0
0	0	3	0	0	0	0	0	0	0
0	0	0	4	0	0	0	0	0	0
0	0	0	0	5	0	0	0	0	0
0	0	0	0	0	6	0	0	0	0
0	0	0	0	0	0	7	0	0	0
0	0	0	0	0	0	0	8	0	0
0	0	0	0	0	0	0	0	9	0
0	0	0	0	0	0	0	0	0	10

```
A = B * D * inv(B)
```

```
A = 10x10
```

4.8189	22.1378	-5.7572	-5.9289	-18.0142	-15.0008	14.1278	15.4067	...	...
-3.3964	33.5466	-4.3981	-3.7181	-19.9303	-24.0164	19.9629	21.5480	...	...
-1.1064	7.2337	1.5880	0.0273	-3.5154	-6.5353	4.8015	6.6623	...	...
-0.5201	31.4656	-4.5998	-2.0321	-25.1948	-25.8689	23.0274	22.1091	...	...
-4.0563	7.6786	-0.2382	2.6131	2.7278	-9.0648	6.6206	7.2083	...	...
-1.5562	20.5415	-3.6837	-2.2358	-13.8276	-14.2622	16.0229	17.2523	...	...
-1.5828	3.9058	-0.4863	3.8136	0.6124	-9.1835	11.4468	7.4840	...	...
-2.4122	5.3808	1.1126	0.5131	-3.1709	-4.7916	2.3517	12.5092	...	...
-1.8147	7.3761	-2.4287	-1.5177	-4.5267	-3.2570	2.0883	8.9527	...	...
-1.9236	6.8626	5.1774	2.4866	-5.1123	-11.9238	5.5819	12.7089	...	...

6) Create a 4×4 non-diagonal matrix with eigenvalues as 1,2,3,4. Find the following,

```
clc;
% Create a manual 4 * 4 non diagonal matrix
V = rand(4);
D = diag([1, 2, 3, 4]);
A = V * D * inv(V)
```

```
A = 4x4
```

-0.0544	2.2788	-3.1382	3.8025
-2.3754	4.3959	-2.0437	2.6763
-3.3213	3.3999	-0.8458	3.5764
-2.4901	2.4783	-3.1874	6.5043

```
I = eye(4);
```

a) Rank of $A-I$, $A-2I$, $A-3I$, $A-4I$

```
AI = A - 1*I;
A2I = A - 2*I;
```

```
A3I = A - 3*I;
A4I = A - 4*I;
R_1I = rank(AI)
```

```
R_1I =
3
```

```
R_2I = rank(A2I)
```

```
R_2I =
3
```

```
R_3I = rank(A3I)
```

```
R_3I =
3
```

```
R_4I = rank(A4I)
```

```
R_4I =
3
```

b) Rank of (A-1I)* (A-2I)

```
R_P = rank(AI * A2I)
```

```
R_P =
2
```

c) Rank of (A-1I)* (A-2I)*(A-3I)

```
R_P2 = rank(AI * A2I * A3I)
```

```
R_P2 =
1
```

d) Rank of (A-1I)* (A-2I)*(A-3I)*(A-4I)

```
R_R = rank(AI * A2I * A3I * A4I)
```

```
R_R =
4
```

7) Create 4x 4 matrices with

```
clc
Z= randi([0,9],4,4)*randi([0,9],4,4);
Z1=rank(Z)
```

```
Z1 =
4
```

```
P=inv(Z);
```

(i) only one eigenvalue as 0,

```
A1= Z *diag([1 2 3 0])*P
```

```
A1 = 4x4
12.9148 -2.0618 2.3225 -15.9487
22.8754 -1.5834 -7.0583 -23.7438
4.0027 -0.9996 5.3708 -7.8198
9.5706 -1.5286 0.2793 -10.7022
```

```
eig(A1)
```

```
ans = 4x1
3.0000
0.0000
2.0000
1.0000
```

```
Z2=rank(A1)
```

```
Z2 =
3
```

(ii) only two eigenvalues as 0,

```
A2= Z *diag([1 2 0 0])*P
```

```
A2 = 4x4
48.7973 -4.6018 -38.7499 -29.8728
42.7828 -2.9926 -29.8451 -31.4688
28.8256 -2.7567 -23.0423 -17.4523
32.9188 -3.1814 -26.4458 -19.7624
```

```
eig(A2)
```

```
ans = 4x1
2.0000
1.0000
-0.0000
0.0000
```

```
Z3=rank(A2)
```

```
Z3 =
2
```

(iii) only three eigenvalues as 0.

Find the rank in each case and comment on the change in rank.

```
A3= Z *diag([1 0 0 0])*P
```

```
A3 = 4x4
13.6623 -0.2786 -6.8481 -13.4787
22.1292 -0.4513 -11.0919 -21.8318
7.6971 -0.1570 -3.8581 -7.5937
8.4668 -0.1727 -4.2439 -8.3530
```

```
eig(A3)
```

```
ans = 4x1
1.0000
-0.0000
-0.0000
```

0.0000

```
Z4=rank(A3)
```

```
Z4 =  
1
```

Observation: If we have 3 different 4x4 matrices and each matrix has 1 eigen value as 0, 2 eigen values as 0 and 3 eigen values as 0 then the rank of each of the matrices will reduce to 3, 2 and 1.

8) Create 29×9 matrices each with rank 7 using the concept of Cayley Hamilton Theorem mentioned above. What is the rank of the product of these two matrices.

```
clc  
V1 = randn(9);  
V2 = randn(9);  
  
% Define diagonal matrices with 7 nonzero and 2 zero eigenvalues ( If a matrix has  
% 2 zero eigenvalues, its rank is at most 7)  
D1 = diag([randi([1, 10], 1, 7), 0, 0]);  
D2 = diag([randi([1, 10], 1, 7), 0, 0]);  
  
A = V1 * D1 * inv(V1)
```

```
A = 9x9  
 4.3267 -25.7174 -16.6788 16.0619 -57.9484 -10.7138 -13.3084 -24.1616 ...  
 0.0985 13.4854 2.1246 -3.2556 15.2044 1.6951 -0.9877 5.7958  
 -0.2117 -10.3465 -0.2590 4.4815 -20.9368 -3.1956 -10.7742 -11.4511  
 -0.8750 -16.7497 -12.7046 15.1177 -39.4316 -10.7216 -13.2941 -16.4479  
 -3.1957 -17.0438 -10.9967 11.2348 -31.0178 -10.7932 -8.7728 -13.6203  
 -0.4178 5.0633 6.8395 -5.1340 15.7647 5.2888 4.3997 7.3288  
 0.1541 -2.2798 -3.7748 1.8598 -3.6919 -2.1656 3.5412 -1.0086  
 4.5340 15.3724 11.3444 -11.3049 31.0820 13.0212 12.2303 15.8880  
 5.7389 27.9591 16.4616 -17.5329 52.6094 18.0993 17.6794 23.1035
```

```
B = V2 * D2 * inv(V2)
```

```
B = 9x9  
 6.4148 0.6842 16.0134 -4.4902 1.6712 1.6621 2.5425 4.8908 ...  
 0.4660 3.8184 3.3757 -0.6753 -1.3860 -1.1425 -0.8519 1.9617  
 -1.3081 -0.2517 -3.1359 1.9436 0.0464 0.1632 -1.0967 -2.3828  
 1.1640 -1.1347 -0.7093 2.9050 1.2816 -1.3201 2.3632 -1.6743  
 2.0718 1.2272 8.9394 -2.2966 2.7618 -0.4735 -0.3373 3.1090  
 1.9092 0.0196 8.7049 -2.2134 0.4622 4.8663 2.3082 0.8314  
 -0.8968 -1.2032 -3.6335 0.9178 -0.6102 -0.4838 3.5304 0.1964  
 3.4138 0.3642 4.0142 -2.3235 0.1422 -2.7899 2.5305 4.1727  
 1.5904 1.2026 6.0684 -2.1055 0.0589 0.4065 -0.1680 2.4928
```

```
% The rank of a matrix is equal to the number of nonzero eigenvalues. (rank(AB)  
≤ min(rank(A), rank(B))  
rankA = rank(A)
```

```
rankA =  
7
```

```
rankB = rank(B)
```

```
rankB =  
7
```

```
AB = A * B;  
rankAB = rank(AB)
```

```
rankAB =  
7
```

9) Create a 4×4 matrix Φ , for which $A^4 = 0$ matrix.

```
clc  
% Create 4x4 lower triangle matrix with rank 3 and with eigen values as zero  
B=zeros(4);  
% Setting non zero values in the lower triangle  
B(2,1)=2;  
B(3,1)=4;  
B(3,2)=2;  
B(4,1)=3;  
B(4,2)=4;  
B(4,3)=2;  
A=B
```

```
A = 4x4  
    0    0    0    0  
    2    0    0    0  
    4    2    0    0  
    3    4    2    0
```

```
A_4=A^4
```

```
A_4 = 4x4  
    0    0    0    0  
    0    0    0    0  
    0    0    0    0  
    0    0    0    0
```

10) Using MATLAB generate a 10×5 matrix A of rank 3.

a) Obtain a symmetric matrix $\Phi_1 = \Phi * \Phi^T$ and $\Phi_2 = \Phi^T * \Phi$.

```
clc  
A = randi([0 9], 10, 3)*randi([0 9], 3, 5)
```

```
A = 10x5  
    16    15    44    54    12  
     8    63   106    84    42  
    27    99   153   126    54  
     2    30    58    48    24  
    12    83    82    40    30  
     3    39    66    51    27  
    26    24    61    75    15  
    30    72   135   129    45
```


12	85	89	47	33
18	68	109	91	39

$S1 = A' * A$

$S1 = 5 \times 5$

3250	9618	15647	13689	5313
9618	40914	59766	46596	21492
15647	59766	92773	76080	33054
13689	46596	76080	65109	26739
5313	21492	33054	26739	11889

$S2 = A * A'$

$S2 = 10 \times 10$

5477	10777	16101	5914	7565	6615 ...
10777	24089	35523	13094	18637	14895
16101	35523	52731	19242	27747	21924
5914	13094	19242	7148	9910	8100
7565	18637	27747	9910	16257	11535
6615	14895	21924	8100	11535	9216
7690	15116	22671	8270	10756	9270
15006	31812	47277	17322	23916	19602
8317	20219	30060	10784	17367	12513
11486	25264	37467	13702	19608	15594

b) Find the rank of 1 and 2.

rank(S1)

ans =
3

rank(S2)

ans =
3

c) Find the eigenvalues of 1 and 2. What do you notice about them? Explain.

EvalS1= eig(S1)

EvalS1 = 5x1
10⁵ x
-0.0000
0.0000
0.0057
0.0515
2.0821

EvalS2 = eig(S2)

EvalS2 = 10x1
10⁵ x
-0.0000
-0.0000
0.0000
0.0000
0.0000
0.0000

0.0000
0.0057
0.0515
2.0821

Observation: The non zero eigen values of S_1 and S_2 are the same. Hence S_2 and S_1 are similar.